Assignment 06
Resistively Loaded CS Amplifier

ECEN 222, Spring 2026

University of Nebraska-Lincoln

Overview

This assignment focuses on the analysis and simulation of a common-source (CS) amplifier with resistive load. You will perform theoretical calculations to determine the DC bias point and small-signal gain, then verify your results using Multisim simulation. Understanding CS amplifiers is fundamental to analog circuit design and MOSFET-based amplification.

Submission Requirements

Submit a **single PDF document** containing:

1. Paper-pencil analysis results for V_{GS} and the small-signal gain. Show your work.
2. Screenshot of the schematic.
3. DC operating point simulation results.
4. Screenshots of DC sweep analysis graph (VTC), with cursors showing the slope around the intended DC bias point.
5. Screenshot of transient analysis graph, with cursors showing the output peak-to-peak voltage. Also provide the gain magnitude in writing.
6. Comment on the differences between paper-pencil and simulation results.

Software Requirements

- **Multisim:** Available through the NI Academic Site License. Download from `https://www.ni.com/` or use computers in the lab.

Circuit Description

Consider the Multisim schematic of the common-source (CS) amplifier (similar to Fig. 7.10 in lecture notes) shown below, where the component values and the NMOS model parameters are shown:

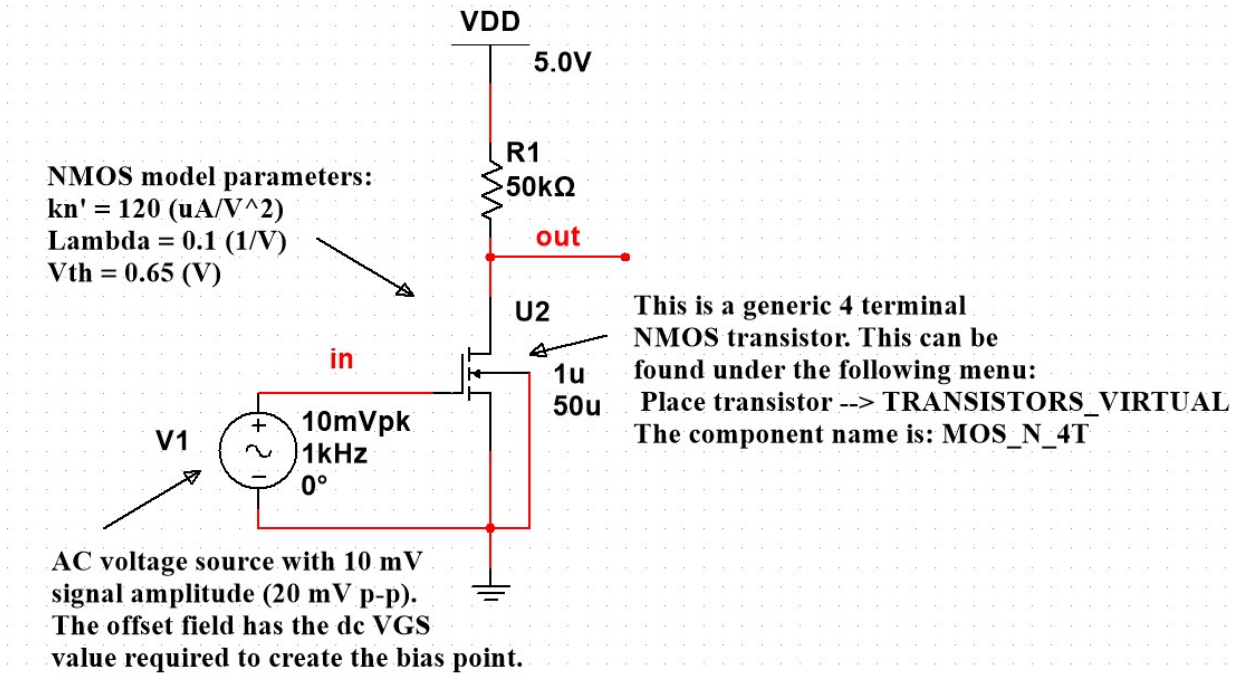

Figure 1: Resistively loaded CS-Amplifier.

In this CS-amplifier, we are using a generic 4-terminal NMOS transistor, which lets us modify its model parameters in accordance with the values shown in the schematic and hence, create a custom NMOS device for a given microelectronics fabrication process.

The NMOS transistor model parameters displayed in the schematic are typical of a 0.5 micron CMOS fabrication technology with a $V_{DD} = 5$ V. Also note in the schematic that the NMOS transistor channel length = 1 micron and channel width = 50 microns.

NMOS Transistor Configuration in Multisim

When properties of the NMOS device are selected, the top window will let you enter the W (channel width) and L (channel length) values. Also, the default boxes for W and L should be unchecked. For units of micron, "u" is utilized, i.e. 50u and 1u for the width and length, respectively.

The model window is displayed by pressing the "edit model" button (see Figure 2). This window will let you enter the device model parameters V_{th}, k'_n, and λ, where:

- $V_{th} \leftrightarrow$ VTO
- $k'_n \leftrightarrow$ KP
- $\lambda \leftrightarrow$ LAMBDA

Finally, the "device model level" shown in the top row of the model window should be set to "MOS 1 (level 1)" from the pull down menu. The default boxes for the entered values should be unchecked as well.

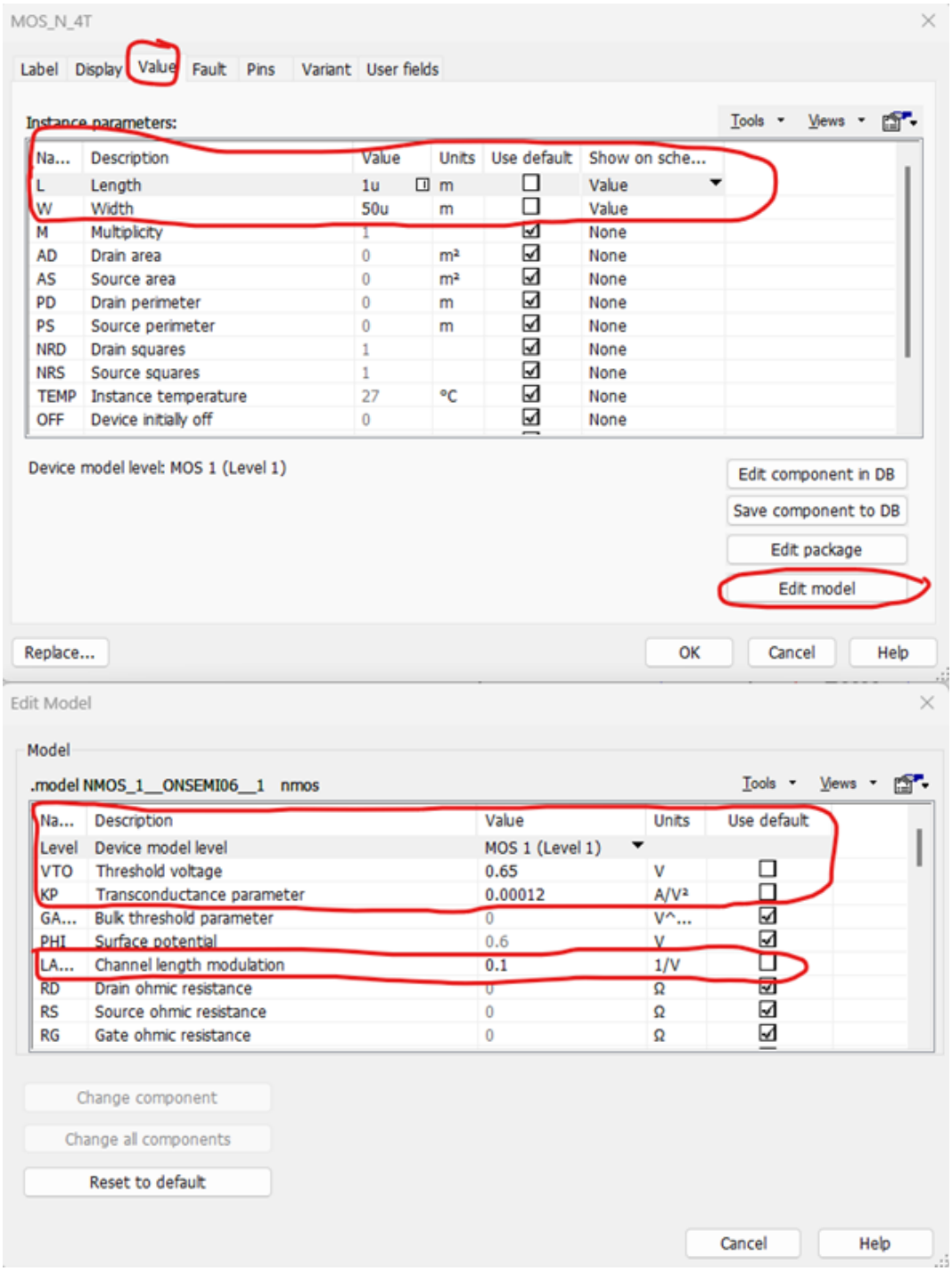

Figure 2: NMOS transistor model parameter configuration window in Multisim.

Question 1: Paper Pencil Calculations [50 points]

For the CS amplifier schematic shown, with $V_{DD} = 5$ V and $R_1 = 50$ kΩ, where the NMOS parameters are given as:

- $k_n' = 120\ \mu\text{A/V}^2$

- $\lambda = 0.1\ \mathrm{V}^{-1}$
- $V_{th} = 0.65$ V
- Channel length: $L = 1\ \mu$m
- Channel width: $W = 50\ \mu$m

Part i) DC Bias Point Calculation [25 points]

Determine what DC V_{GS} value must be used (this is the DC offset voltage of the AC voltage source) for biasing the circuit such that the output node sits at 2.5 V. Include the channel length modulation term in the MOSFET current equation.

Part ii) Small-Signal Gain Calculation [25 points]

For this bias setting (that is, once the V_{GS} and the corresponding I_{DS} values are determined), calculate the small signal gain of this CS amplifier. Include r_o (the output resistance due to channel length modulation effect) in the gain calculation.

Question 2: Multisim Simulation [50 points]

Part i) Schematic Creation [10 points]

Create the schematic of CS amplifier as shown on the first page. Use an AC voltage source with:

- Peak amplitude: $V_{pk} = 10$ mV
- Frequency: $f = 1$ kHz
- DC offset voltage: calculated in Q1, part i)

Also configure the generic 4-terminal NMOS transistor per values indicated on the second page. Tie the bulk terminal of the NMOS device to the ground node.

Part ii) DC Operating Point Analysis [10 points]

Run "DC operating point" analysis and observe the DC value of V_{out}.

Part iii) DC Sweep Analysis [15 points]

Run "DC Sweep" analysis and obtain the VTC of the amplifier. From this VTC, measure the slope around the intended DC bias point using the measurement cursors on the graph displayed. This slope will be an indicator of the small-signal gain around the bias point.

Part iv) Transient Analysis [15 points]

Run "Transient" analysis for 0.002 seconds (two full cycles of the 1 kHz sine wave) and observe the time-domain signal $V_{out}(t)$. Measure the peak-to-peak voltage of the output waveform and divide it by the 20 mV peak-to-peak input voltage magnitude. This will result in the CS amplifier gain magnitude obtained via transient analysis.